MILITARY SPECIFICATION SHEET

CABLE ASSEMBLIES, NONPRESSURE PROOF, FIBER OPTIC, METRIC, SINGLE BUNDLE, 6-POSITION MIL-C-83526 CONNECTORS AND DOD-C-85045 CABLES

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49292.

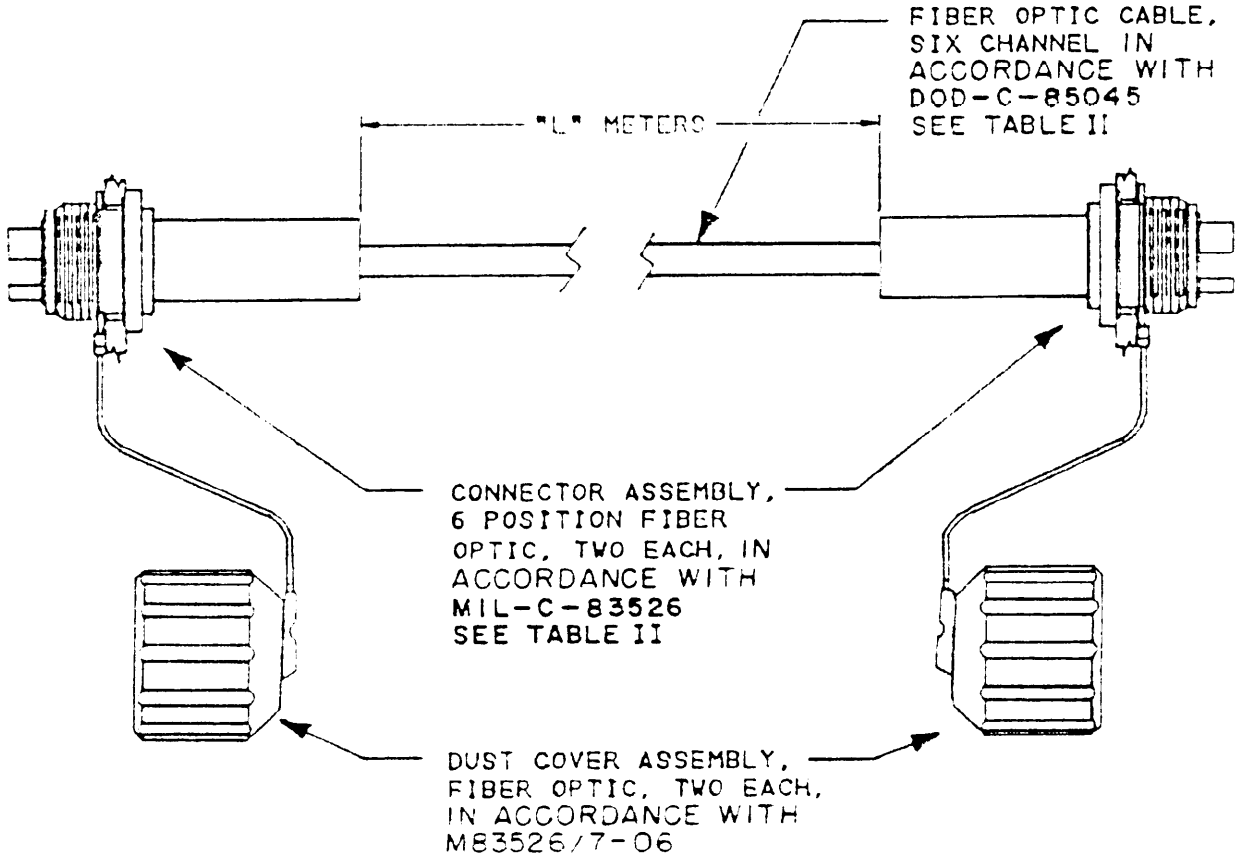


FIGURE 1. Cable harness assembly, complete with dust cover.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Color: Colors for coding of fibers within the cable harness shall be as shown in table I.

TABLE I. Color coded termination configuration.

Connector terminus position	Buffer color	Connector terminus position
1	Blue	4
2	Orange	5
3	Green	6
4	Brown	1
5	Gray	2
6	White	3

Component parts:

Connectors shall be in accordance with the following:

MIL-83526, two required (see table II).

Dust cover assembly: M83526/7-06, two required.

Cable: DOD-C-85045 (see table II).

Pin termini: In accordance with MIL-T-29504/1, three required (positions 2, 4, and 6 of MIL-C-83526 connector) (see table II).

Socket termini: In accordance with MIL-T-29504/2, three required (positions 1, 3, and 5 of MIL-C-83526 connector) (see table II).

Assembly: In accordance with figure 1, and tables I and II.

Length: In accordance with table II.

Weight: In accordance with table II.

Performance:

Optical requirements:

Insertion loss (maximum assembly attenuation): See table II. Test method B1.

Bandwidth: $\geq$ 400 MHz-km at 1290 nm, ±10 nm.

Crosstalk: $\geq$ 50 dB below the active channel.

Water pressure: Applicable.

Temperature range:

Operating: -40°C to +85°C.

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Storage: -55°C to +110°C.

Life test: Life test shall be performed at +85°C ±5°C.

Part or Identifying Number (PIN): PINs shall be in accordance with table II.

TABLE II. PINs, cable length, assembly attenuation, and weight.

Military PIN M49292/2-	Cable length "L" (meters)	Maximum assembly attenuation (dB)		Maximum assembly weight (kg)
		At 820-890 nanometers	At 1100-1300 nanometers	
	Fiber size: 50/125 micron diameter Cable PIN: D85045/3-B6B Connector PIN: M83526/4-06A1A			
01	1000 ±10	8	5	90.0
02	500 ±5	5	3.5	45.2
03	300 ±3	3.8	2.9	27.3
04	200 ±3	3.2	2.6	18.3
05	100 ±3	2.6	2.3	9.4
06	50 ±2	2.3	2.2	4.9
07	25 ±1	2.2	2.1	2.7
08	10 ±1	2.1	2.1	1.3
	Fiber size: 100/140 micron diameter Cable PIN: D85045/3-C6B Connector PIN: M83526/4-06A1B			
09	1000 ±10	7.5	4.5	90.0
10	500 ±5	4.5	3.0	45.2
11	300 ±3	3.3	2.4	27.3
12	200 ±3	2.7	2.1	18.3
13	100 ±3	2.1	1.8	9.4
14	50 ±2	1.8	1.7	4.9
15	25 ±1	1.7	1.6	2.7
16	10 ±1	1.6	1.6	1.3

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SC
Navy - AS, CG, EC, MC
Air Force - 99
DLA - ES

User activities:
Army - AV
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6020-0002-2)

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